IGCSE 0607 Practice Questions — E10.1 — Combined Questions & Markscheme

Classifying statistical data

Total Questions: 9 — **Total Marks:** 60 Marks

Table of Contents & Quick Navigation

- **E10.1** Classifying statistical data
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

---

1 70 students are asked to choose their favourite sport from Football, Tennis, or Basketball.
The incomplete two-way table shows some information about their choices.

(a) Complete the two-way table.

	Football	**Tennis**	**Basketball**	**Total**
Boys	15		12	
Girls		14		30
Total	23		20	70

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Tennis.

..... **[1]**

2 50 students each choose one sport to play: Tennis, Football, or Swimming.
The students are classified as Boys or Girls.

(a) Complete the two-way table to show this information.

	Tennis	**Football**	**Swimming**	**Total**
Boys	7			28
Girls	8		6	
Total		20		50

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy who chose Football.

..... **[1]**

3 80 students were asked to choose their favourite sport from Football, Basketball, and Tennis.

- There are 35 girls in total.
- 38 students chose Football, of which 24 were boys.
- 12 girls chose Basketball.
- 20 students chose Tennis.

(a) Complete the two-way table.

	Football	**Basketball**	**Tennis**	**Total**
Boys				
Girls				
Total				

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Basketball.

..... **[1]**

(c) A girl is chosen at random.
Find the probability that her favourite sport is Tennis.

..... **[2]**

4 A survey of adults recorded their preferred hot drink, either Tea or Coffee.
Let x be the number of males who prefer Coffee.
The number of males who prefer Tea is $2x + 10$.
The number of females who prefer Coffee is $x + 5$.
The number of females who prefer Tea is $3x - 11$.

(a) Write down an expression, in terms of x, for the total number of adults.
Give your answer in its simplest form.

..... **[2]**

(b) The survey included 130 adults.
Find the value of x.

$x =$ ..... **[2]**

(c) Complete the two-way table.

	Coffee	**Tea**	**Total**
Male			
Female			
Total			

[3]

(d) One of the adults who prefers Coffee is chosen at random.
Find the probability that this adult is female.

..... **[2]**

5 A group of 120 adults were asked whether they prefer Tea or Coffee.

- 55 of the adults are women.
- 70 of the adults prefer Tea.
- 25 of the men prefer Coffee.

(a) Complete the two-way table to show this information.

	Tea	**Coffee**	**Total**
Men			
Women			
Total			

[3]

(b) Work out the number of women who prefer Tea as a percentage of the total number of adults.

..... % **[2]**

6 In a school of 400 students, the ratio of juniors to seniors is 3 : 2.
Each student travels to school by Bus, by Car, or Walks.

- 45% of the 400 students take the Bus.
- Of the seniors, 30 take the Car and 50 Walk.
- There are 80 juniors who take the Car.

(a) Complete the two-way table to classify this data.

	Bus	**Car**	**Walk**	**Total**
Juniors				
Seniors				
Total				400

[3]

(b) A student is chosen at random.
Find the probability that they are a junior who walks.

..... **[1]**

(c) Two seniors are chosen at random.
Find the probability that they both take the bus.

..... **[2]**

7 80 students travel to school either by Bus or by Walking.
The two-way table shows the number of boys and girls using each method of transport, in terms of x.

	Bus	**Walk**	**Total**
Boys	x	$2x$	
Girls	$x + 5$	$3x - 2$	
Total			80

(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Complete the two-way table with the number of students in each category.

	Bus	**Walk**	**Total**
Boys			
Girls			
Total			80

[2]

(c) A student is chosen at random from those who walk to school.
Find the probability that this student is a girl.

..... **[2]**

8 A company has 300 employees. Each employee is either a Manager, an Office Worker, or a Technician.
Each employee works either Full-Time or Part-Time.

- The ratio of Managers : Office Workers : Technicians is 2 : 5 : 3.
- 20% of the Managers work Part-Time.
- 40% of the Office Workers work Part-Time.
- The number of Part-Time Technicians is equal to the number of Part-Time Managers.

(a) Complete the two-way table for this information.

	Full-Time	**Part-Time**	**Total**
Managers			
Office Workers			
Technicians			
Total			

[4]

(b) A Part-Time employee is chosen at random.
Find the probability that they are an Office Worker.

..... **[2]**

(c) Two Managers are chosen at random.
Find the probability that exactly one of them works Part-Time.

..... **[3]**

9 A group of 120 students are classified by gender (Boys, Girls) and handedness (Left-handed, Right-handed).
The two-way table shows the number of students in each category in terms of x and y.
Two students are chosen at random from the 120 students.
The probability that both students are left-handed is $\frac{3}{34}$.

	Left-handed	**Right-handed**
Boys	x	y
Girls	$2x$	$y - 6$

(a) Show that $3x^2 - x - 420 = 0$.

[4]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) A right-handed student is chosen at random.
Find the probability that the student is a girl.

..... **[2]**

1 **(a)** Boys Total = 40 and Girls Football = 8 **[B1]**

Girls Basketball = 8 and Total Tennis = 27 **[B1]**

Boys Tennis = 13 **[B1]**

(b) $\frac{13}{70}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Tennis Total = 15, Girls Total = 22 **[B1]**

Boys Football = 12, Boys Swimming = 9 **[B1]**

Girls Football = 8 **[B1]**

(b) $\frac{12}{50}$ or $\frac{6}{25}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** 6 or more correct missing entries **[B3]**

4 or 5 correct missing entries **[B2]**

2 or 3 correct missing entries **[B1]**

(Correct Boys row: 24, 10, 11, 45)

(Correct Girls row: 14, 12, 9, 35)

(Correct Total row: 38, 22, 20, 80)

(b) $\frac{10}{80}$ or $\frac{1}{8}$ **[B1]**

(c) Denominator of 35 seen **[M1]**

$\frac{9}{35}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** $x + (2x + 10) + (x + 5) + (3x - 11)$ **[M1]**

$7x + 4$ **[A1]**

(b) $7x + 4 = 130$ **[M1]**

$x = 18$ **[A1]**

(c) Correct Male row: 18, 46, 64 **[B1]**

Correct Female row: 23, 43, 66 **[B1]**

Correct Total row: 41, 89, 130 **[B1]**

(d) Denominator of 41 seen **[M1]**

$\frac{23}{41}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

5 **(a)** Total Women = 55, Total Tea = 70, Overall Total = 120 placed correctly **[B1]**

Men Coffee = 25 and Total Coffee = 50 or Men Total = 65 placed correctly **[B1]**

Men Tea = 40, Women Tea = 30, Women Coffee = 25 **[B1]**

(b) $\frac{30}{120} \times 100$ **[M1]**

25 **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** Total Juniors = 240, Total Seniors = 160 **[B1]**

Total Bus = 180, Total Car = 110, Total Walk = 110 **[B1]**

Juniors: Bus = 100, Car = 80, Walk = 60 AND Seniors: Bus = 80, Car = 30, Walk = 50 **[B1]**

(b) $\frac{60}{400}$ or $\frac{3}{20}$ **[B1]**

(c) $\frac{80}{160} \times \frac{79}{159}$ **[M1]**

$\frac{79}{318}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** $x + 2x + (x + 5) + (3x - 2) = 80$ **[M1]**

$7x + 3 = 80$ **[M1]**

$x = 11$ **[A1]**

(b) Boys Bus = 11, Boys Walk = 22, Boys Total = 33 **[B1]**

Girls Bus = 16, Girls Walk = 31, Girls Total = 47 and Column totals (Bus = 27, Walk = 53) correct **[B1]**

(c) Denominator of 53 seen **[M1]**

$\frac{31}{53}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Correct Managers row: 48, 12, 60 **[B1]**

Correct Office Workers row: 90, 60, 150 **[B1]**

Correct Technicians row: 78, 12, 90 **[B1]**

Correct Total row: 216, 84, 300 **[B1]**

(b) Denominator of 84 seen **[M1]**

$\frac{60}{84}$ or $\frac{5}{7}$ **[A1]**

(c) $\frac{12}{60} \times \frac{48}{59}$ **[M1]**

$\frac{12}{60} \times \frac{48}{59} + \frac{48}{60} \times \frac{12}{59}$ or $2 \times \left(\frac{12}{60} \times \frac{48}{59}\right)$ **[M1]**

$\frac{96}{295}$ or 0.325 (0.3254...) **[A1]**

[Back to Question] *[Total Marks: 9]*

9 **(a)** Total number of left-handed students $= 3x$ **[B1]**

$\frac{3x}{120} \times \frac{3x-1}{119} = \frac{3}{34}$ **[M1]**

$\frac{9x^2-3x}{14280} = \frac{3}{34}$ or $3x(3x-1) = 1260$ **[M1]**

$9x^2 - 3x = 1260 \implies 3x^2 - x - 420 = 0$ (with no errors seen) **[A1]**

(b) $(3x-36)(x+11.66...) = 0$ or correct substitution into quadratic formula **[M1]**

$x = 12$ **[A1]**

Total students equation: $x + y + 2x + y - 6 = 120$ **[M1]**

$3(12) + 2y - 6 = 120 \implies 2y = 90 \implies y = 45$ **[A1]**

(c) Total right-handed $= 2y - 6 = 2(45) - 6 = 84$ OR Right-handed girls $= 45 - 6 = 39$ **[M1]**

$\frac{39}{84}$ or $\frac{13}{28}$ **[A1]**

[Back to Question] *[Total Marks: 10]*